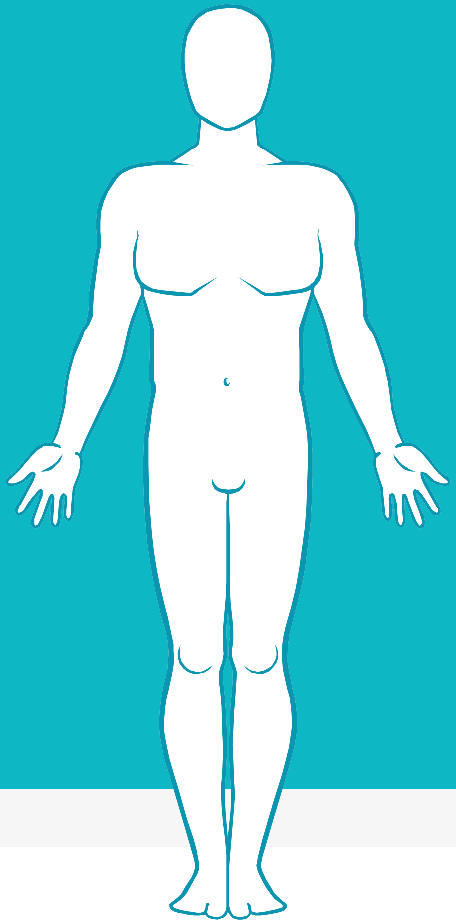
Dasar Sistem Pengaturan dan Laboratorium

Automatic Control Systems in Biomedical Engineering

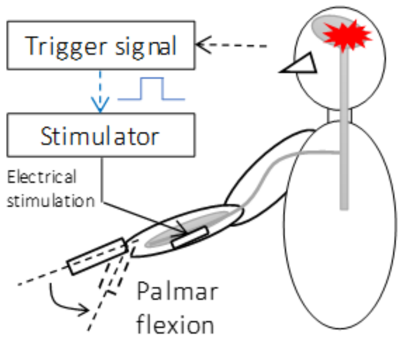
Fauzan Arrofiqi



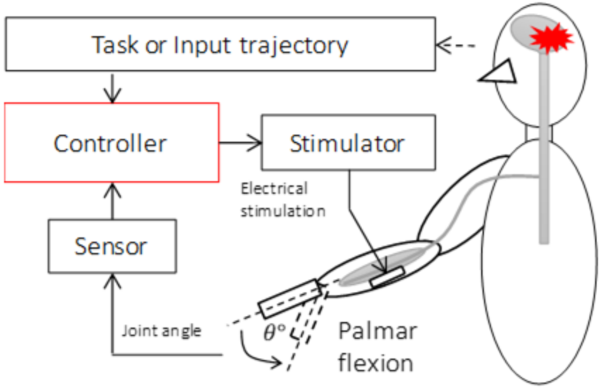
FES Controller for Wrist Joint Control (Computer Simulation)

- **Functional Electrical Stimulation (FES)**
  - Effectively to restore paralyzed motor function caused by spinal cord injury or cerebrovascular disease
  - FES in clinical practice → Trigger method
- **Closed-Loop FES control system**
  - To restore desired functional movements in daily life such as grasping and walking
  - An appropriate FES controller
- **The difficulty in controlling limb movement by FES**
  - High nonlinearity and large time delays in the response of the electrically stimulated musculoskeletal system

FES Trigger method

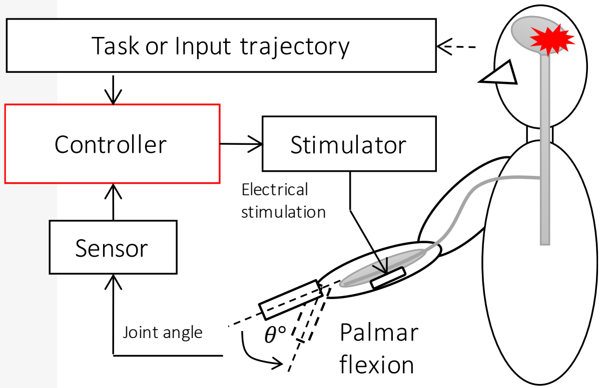


Closed-Loop FES Control System

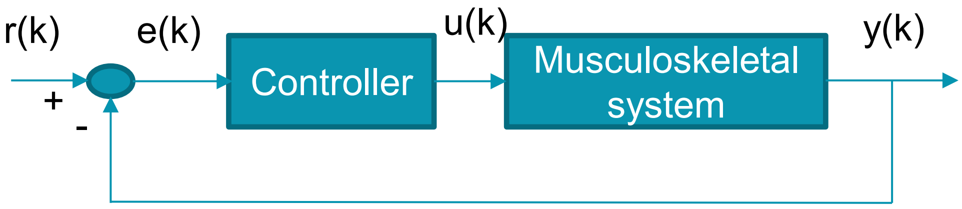


FES Controller for Wrist Joint Control (Computer Simulation) 2

Closed-Loop FES Control System



Computer Simulation

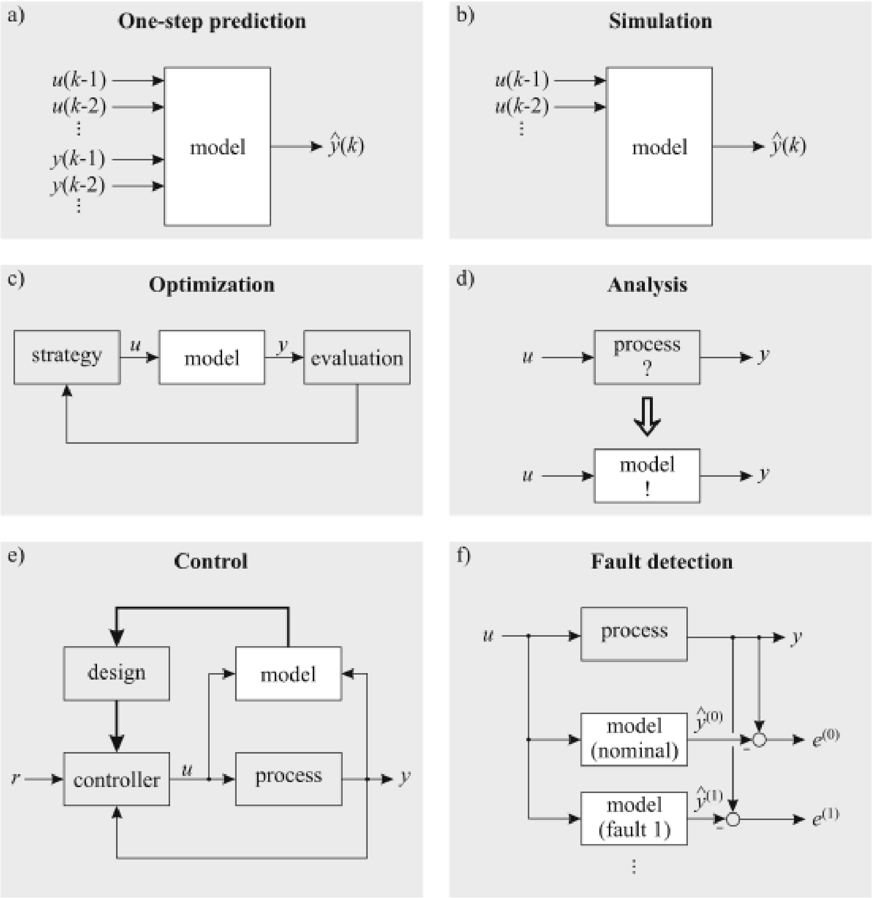


Real system?

Musculoskeletal System

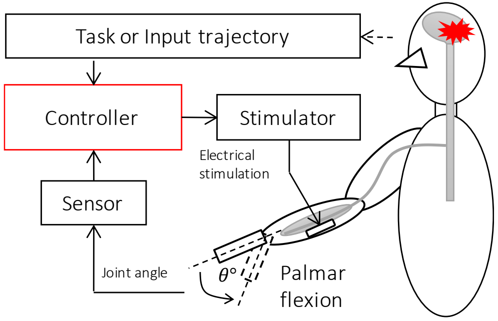
- **Computer simulation**
  - To evaluate the performance of controller
  - Model of plant is required

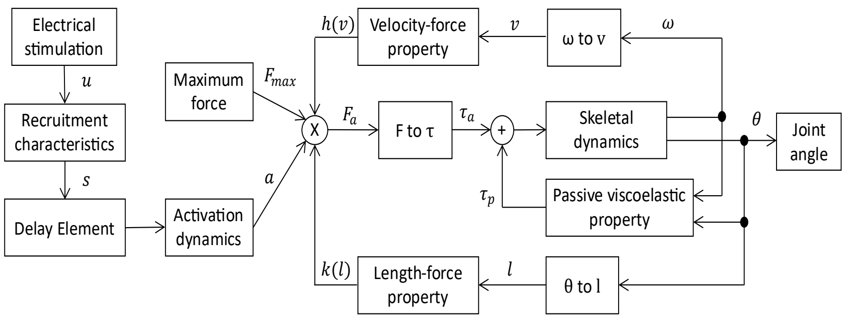
	White box	Gray box	Black box
Information sources	first principles insights	qualitative knowledge (rules) some insights + some data	experiments data
Features	good extrapolation good understanding high reliability scalable		short development time little domain expertise req. can be used in addition also for not understood proc.
Drawbacks	time consuming detailed domain expertise req. knowledge restricts accuracy only for well understood proc.		no reliable extrapolation not scalable data restricts accuracy little understanding
Application areas	planning, construction, design rather simple processes		only for existing processes rather complex processes



Musculoskeletal System (2)

Closed-Loop FES Control System





	White box	Gray box	Black box
Information sources	first principles insights	qualitative knowledge (rules) some insights + some data	experiments data
Features	good extrapolation good understanding high reliability scalable		short development time little domain expertise req. can be used in addition also for not understood proc.
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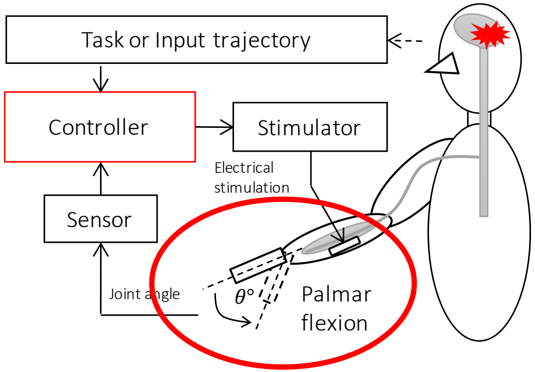
$$\frac{d^2y}{dt^2} + a_1\frac{dy}{dt} + a_0 y(t) = b_0 u(t)$$

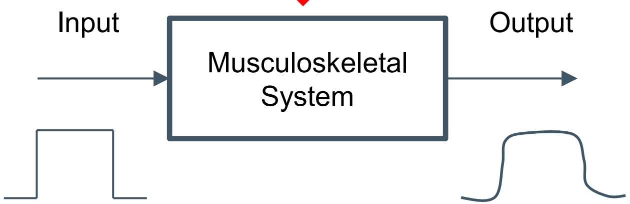
$$G(s) = \frac{K}{\tau s + 1}$$

Systems Identification

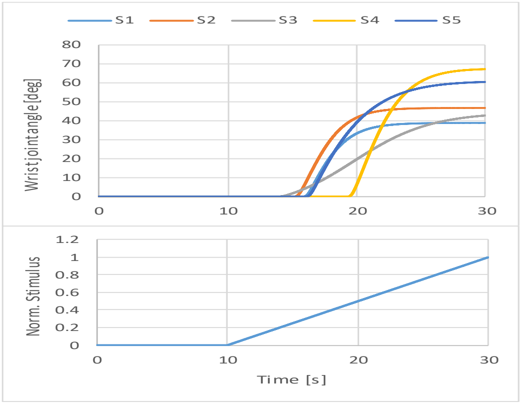
Black Box

Closed-Loop FES Control System

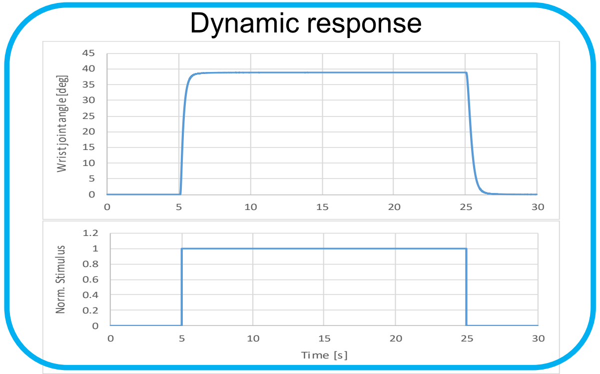




Static response



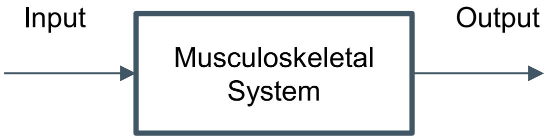
Dynamic response



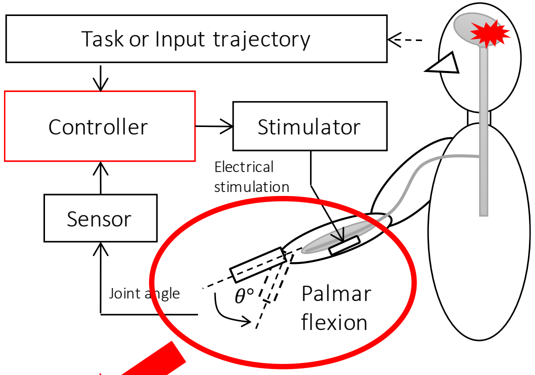
$$G(s) = \frac{K}{\tau s + 1}$$

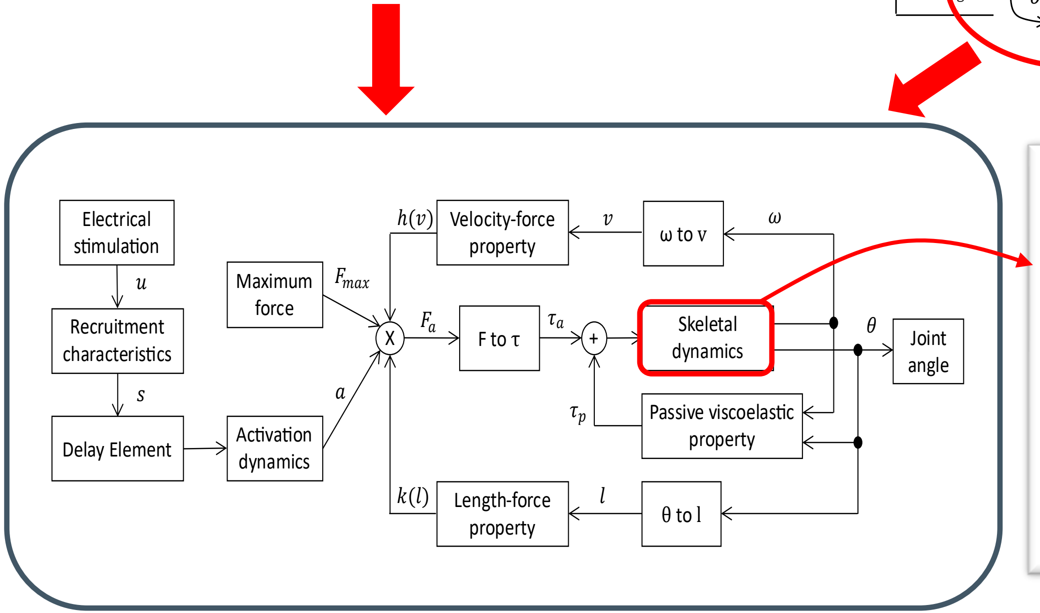
Systems Identification

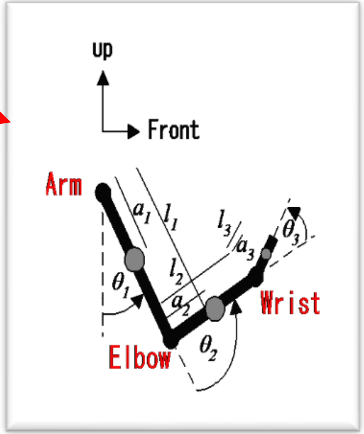
White Box



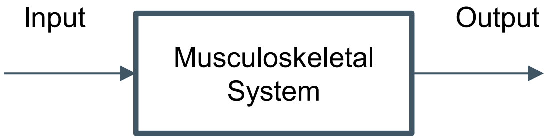
Closed-Loop FES Control System

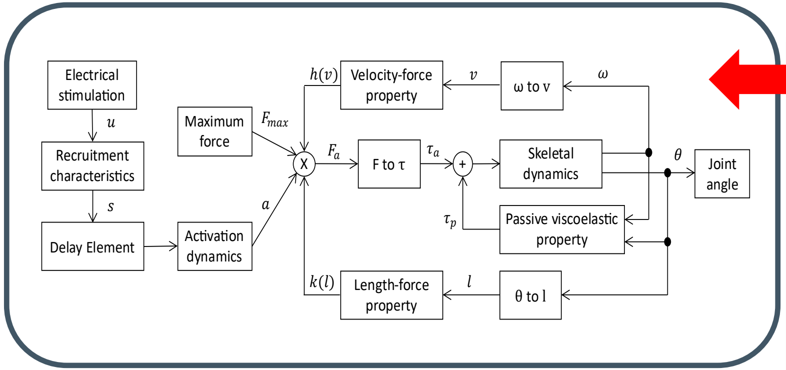




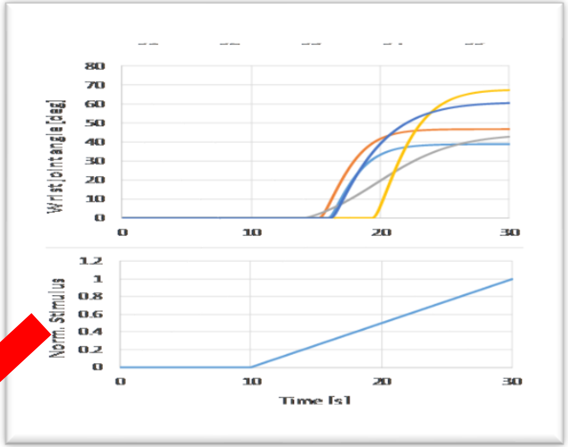


Gray Box

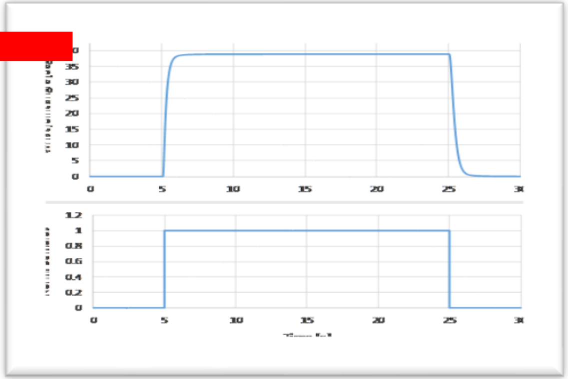




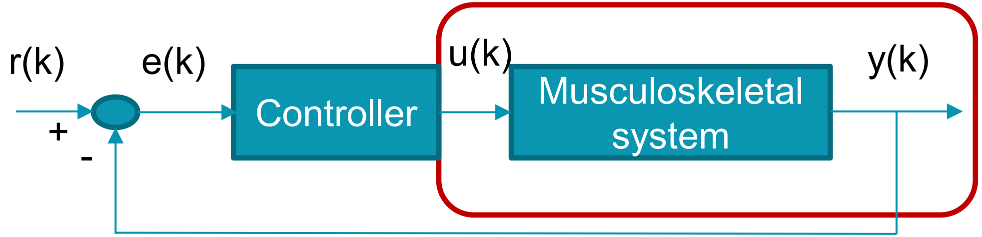
Static response

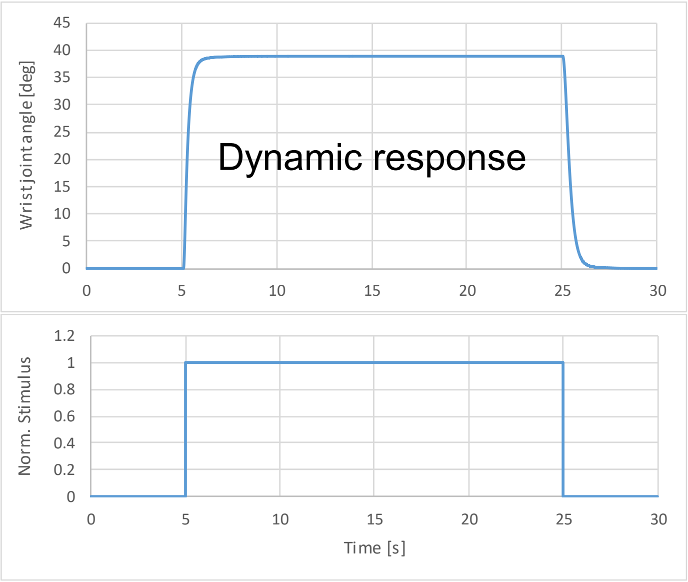


Dynamic response



Musculoskeletal System Model Using BlackBox Method





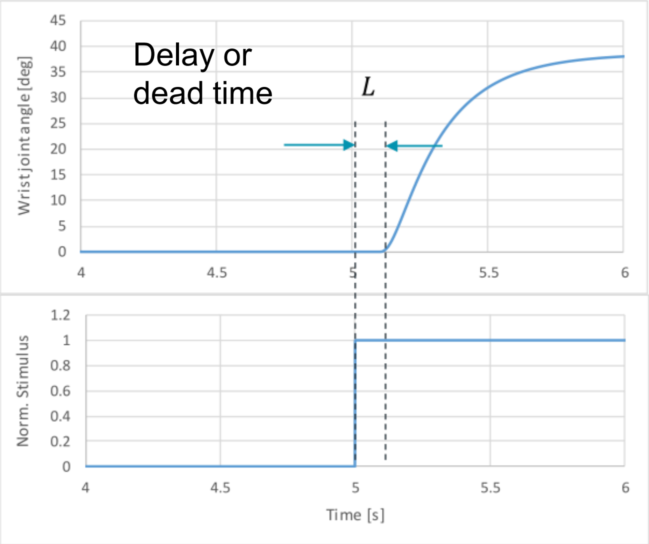
Systems Identification

$$G(s) = \frac{K}{\tau s + 1}$$

First Order Plus Dead Time (FOPDT) Systems

- Time-Delayed Systems are those exhibiting a reaction time, which is the most common and realistic situation
- Biological processes: insulin metabolism, muscle stimulation, etc…, may have a significant delay
- The 2nd order systems can be approximated using FOPDT model

Dynamic response



Delay or dead time unit

$$G(s) = \frac{Y(s)}{U(s)} = \frac{Ke^{-Ls}}{\tau s + 1} \Longrightarrow G(z) = \frac{Y(z)}{U(z)} \Longrightarrow y(n) = G(z)u(n)$$

Dead Time Approximation Method

FOPDT Systems

Delay unit

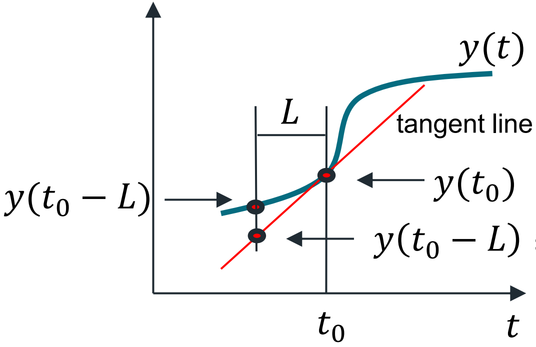
$$G(s) = \frac{Ke^{-Ls}}{\tau s + 1} \rightarrow G(z) = G(s)\big|_{s=\frac{2}{T}\frac{z-1}{z+1}}$$

1. Taylor expansion:

$$e^{-Ls} \cong 1 - Ls$$

Bilinear transformation:

$$e^{-L\frac{2}{T}\frac{z-1}{z+1}} \cong 1 - L\frac{2}{T}\frac{z-1}{z+1}$$



$$y(t_0 - L) \cong y(t_0) - L\frac{dy(t)}{dt_0}$$

The accuracy depends on the dead time

Bilinear transformation:

$$e^{-L\frac{2}{T}\frac{z-1}{z+1}} \cong 1 - L\frac{2}{T}\frac{z-1}{z+1}$$

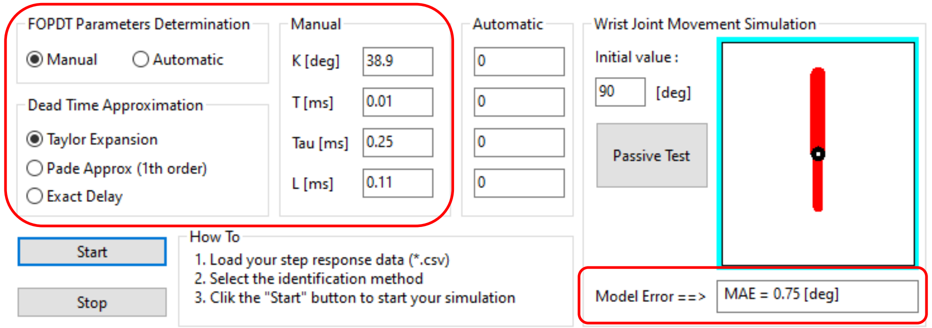
$$G(z) = \frac{K\left(1 - \frac{2L}{T}\frac{z-1}{z+1}\right)}{\left(\frac{2\tau}{T}\frac{z-1}{z+1} + 1\right)}$$

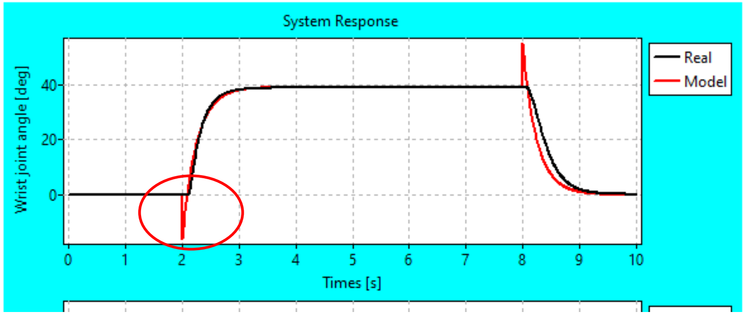
$$G(s) = \frac{Ke^{-Ls}}{\tau s + 1} \rightarrow G(z) = G(s)\Big|_{s = \frac{2}{T}\frac{z-1}{z+1}}$$

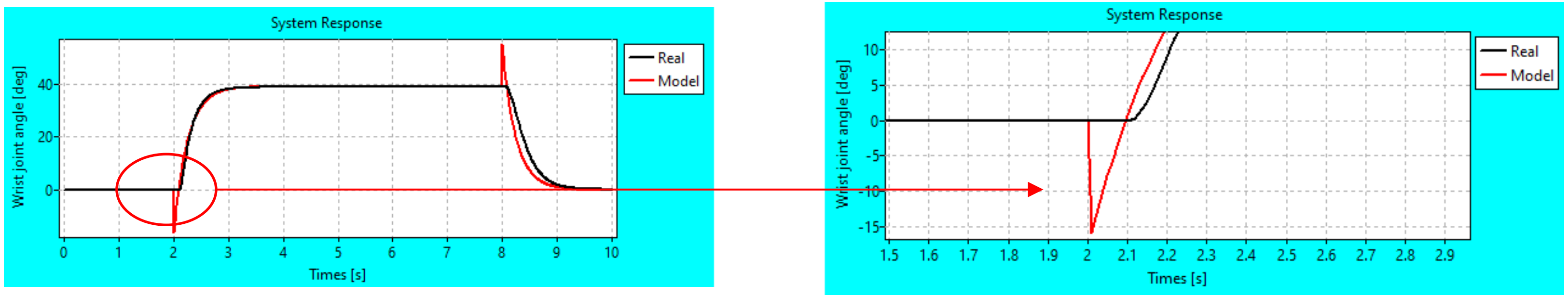
$$y(n) = -cy(n-1) + K\left(au(n) + bu(n-1)\right)$$

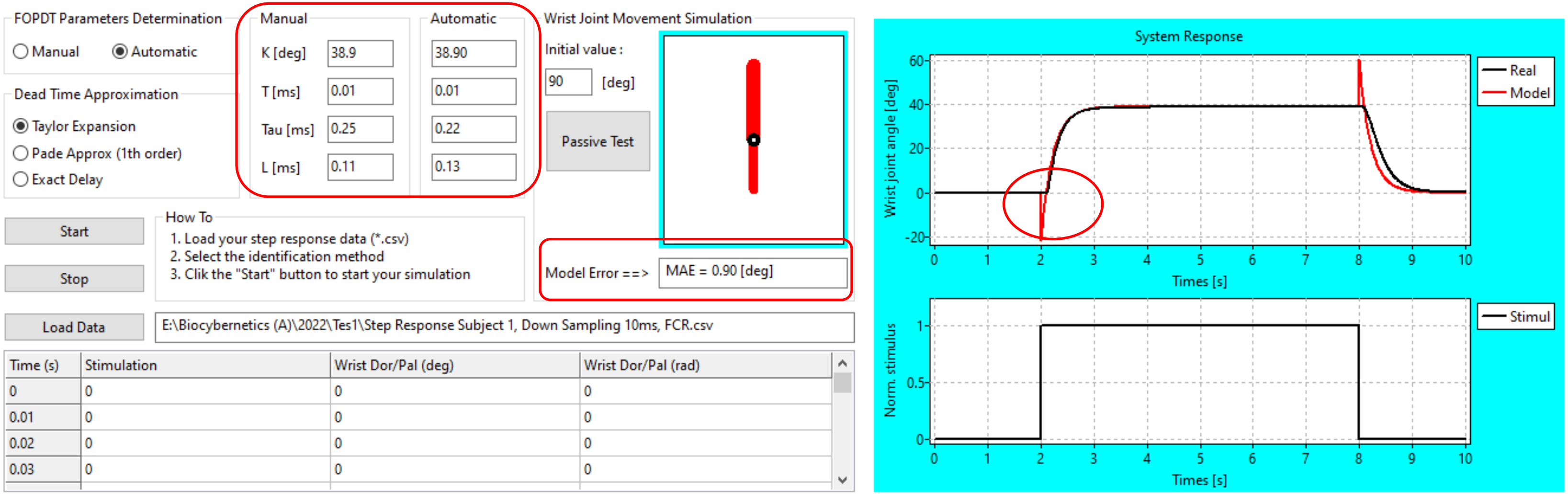
$$a = \frac{(T-2L)}{(T+2\tau)} \quad b = \frac{(T+2L)}{(T+2\tau)} \quad c = \frac{(T-2\tau)}{(T+2\tau)}$$

$$G(s) = \frac{38.9e^{-0.11s}}{0.25s + 1}$$

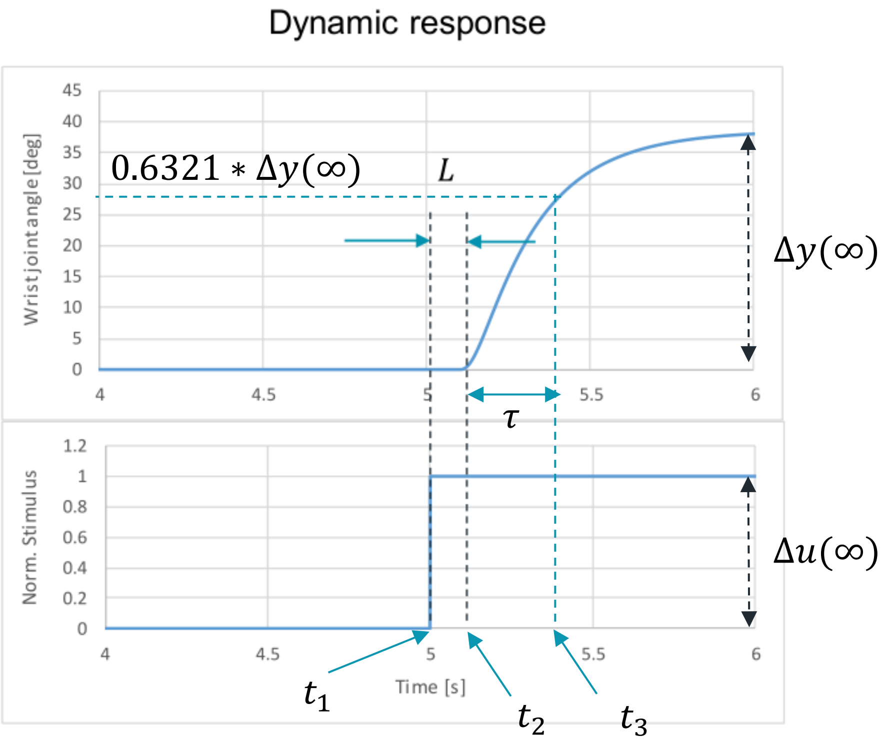





System Response
Real
Model
Wrist joint angle [deg]
Times [s]
System Response
Real
Model
Wrist joint angle [deg]
Times [s]


FOPDT Parameters Determination
Manual
Automatic
Dead Time Approximation
Taylor Expansion
Pade Approx (1th order)
Exact Delay
Manual
Automatic
K [deg] 38.9 38.90
T [ms] 0.01 0.01
Tau [ms] 0.25 0.22
L [ms] 0.11 0.13
Wrist Joint Movement Simulation
Initial value :
90 [deg]
Passive Test
Start
Stop
Load Data
How To
1. Load your step response data (*.csv)
2. Select the identification method
3. Clik the "Start" button to start your simulation
Model Error ==> MAE = 0.90 [deg]
E:\Biocybernetics (A)\2022\Tes1\Step Response Subject 1, Down Sampling 10ms, FCR.csv
Time (s) Stimulation Wrist Dor/Pal (deg) Wrist Dor/Pal (rad)
0 0 0 0
0.01 0 0 0
0.02 0 0 0
0.03 0 0 0
System Response
Real
Model
Wrist joint angle [deg]
Times [s]
Stimul
Norm. stimulus
Times [s]

Automatic determination of FOPDT parameters



$$G(s) = \frac{Ke^{-Ls}}{\tau s + 1}$$

$$K = \frac{\Delta y(\infty)}{\Delta u(\infty)}$$

$$L = t_2 - t_1$$

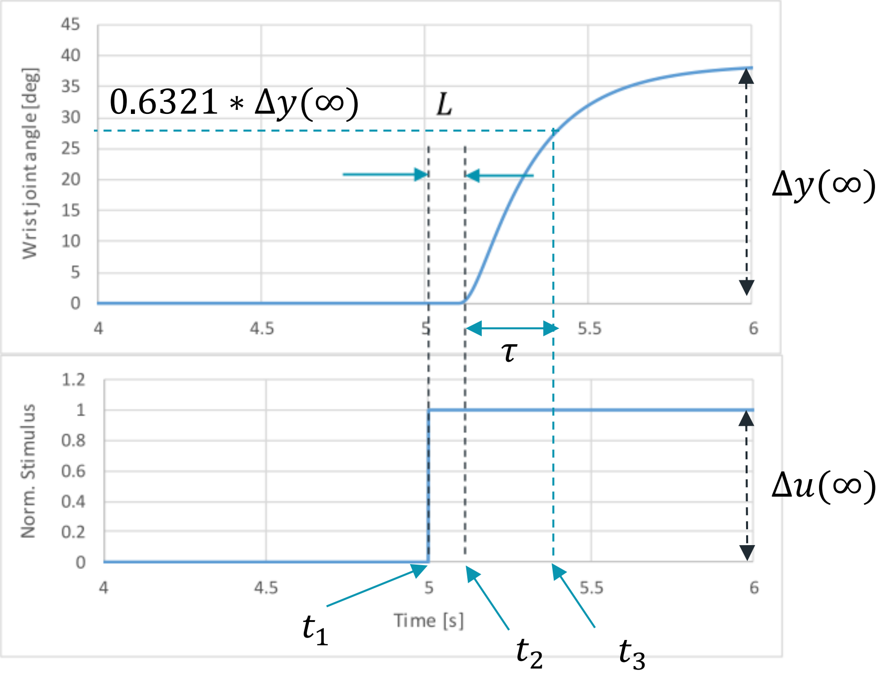
$$\tau = t_3 - t_2$$

Systems Identification

- Ziegler-Nichols method
- Smith's method
- Sundaresan and Krishnaswamy method

Automatic determination of FOPDT parameters

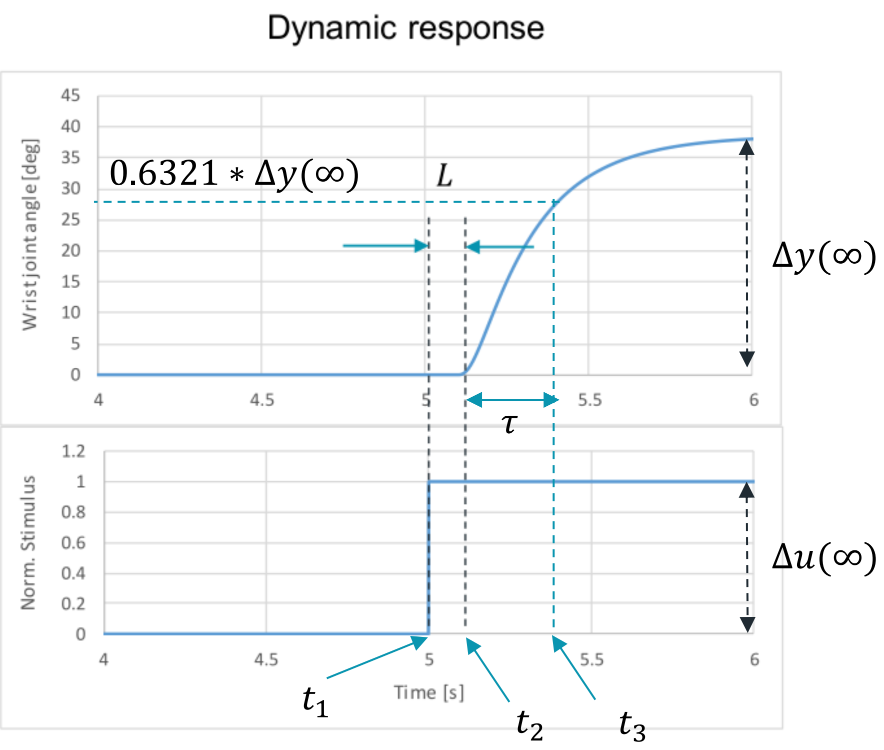
Dynamic response



$$K = \frac{\Delta y(\infty)}{\Delta u(\infty)}$$

```
temp1:=0;
temp2:=0;
for n:=1 to 1001 do
begin
    t[n-1]:=StrToFloat(stringgrid1.Cells[0,n]);
    Stim[n-1]:=StrToFloat(stringgrid1.Cells[1,n]);
    Theta_deg[n-1]:=StrToFloat(stringgrid1.Cells[2,n]);
    //Theta_rad[n-1]:=StrToFloat(stringgrid1.Cells[3,n]);
    temp1:=max(temp1,Theta_deg[n-1]);
    temp2:=max(temp2,Stim[n-1]);
end;
delta_y:=temp1;
delta_u:=temp2;
K:=delta_y/delta_u;
Ts:=t[2]-t[1];
end;
```

Automatic determination of FOPDT parameters



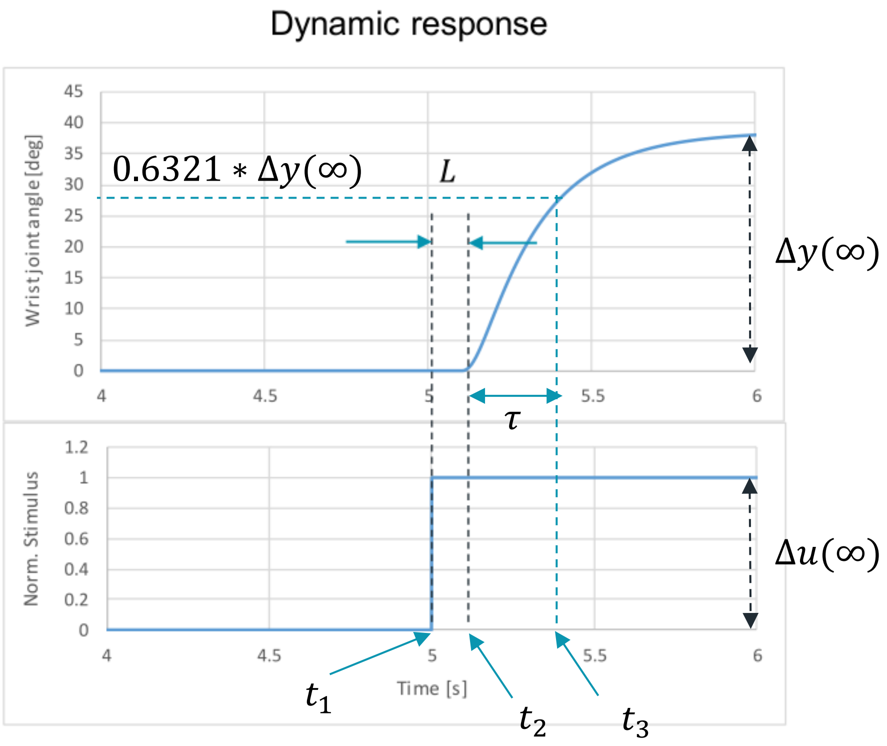
$$L = t_2 - t_1$$

$$\tau = t_3 - t_2$$

```
temp3:=0.00001*delta_u;
for n:=0 to 1000 do
begin
  if temp3<Stim[n] then
  begin
    t1:=n*Ts;
    temp3:=delta_u+1;
  end;
end;
```

```
temp3:=1;//1 deg
for n:=0 to 1000 do
begin
  if temp3<Theta_deg[n] then
  begin
    t2:=n*Ts;
    temp3:=delta_y+100;
  end;
end;
```

Automatic determination of FOPDT parameters



$$L = t_2 - t_1$$

$$\tau = t_3 - t_2$$

```
temp3:=0.6321*delta_y;
for n:=0 to 1000 do
begin
  if temp3<Theta_deg[n] then
  begin
    t3:=n*Ts;
    temp3:=delta_y+100;
  end;
end;
Tau:=t3-t2;
Edit7.Text:=floattostr(Tau);
L:=t2-t1;
Edit8.Text:=floattostr(L);
end;
```

2. Pade approximation:

$$e^{-Ls} \cong \frac{e^{-\frac{L}{2}s}}{e^{\frac{L}{2}s}} \cong \frac{1 - \frac{Ls}{2} + \cdots + \frac{\left(-\frac{Ls}{2}\right)^k}{k!}}{1 + \frac{Ls}{2} + \cdots + \frac{\left(\frac{Ls}{2}\right)^k}{k!}}$$

The first-order Pade approximation:

$$e^{-Ls} \cong \frac{1 - \frac{Ls}{2}}{1 + \frac{Ls}{2}} \cong \frac{2 - Ls}{2 + Ls}$$

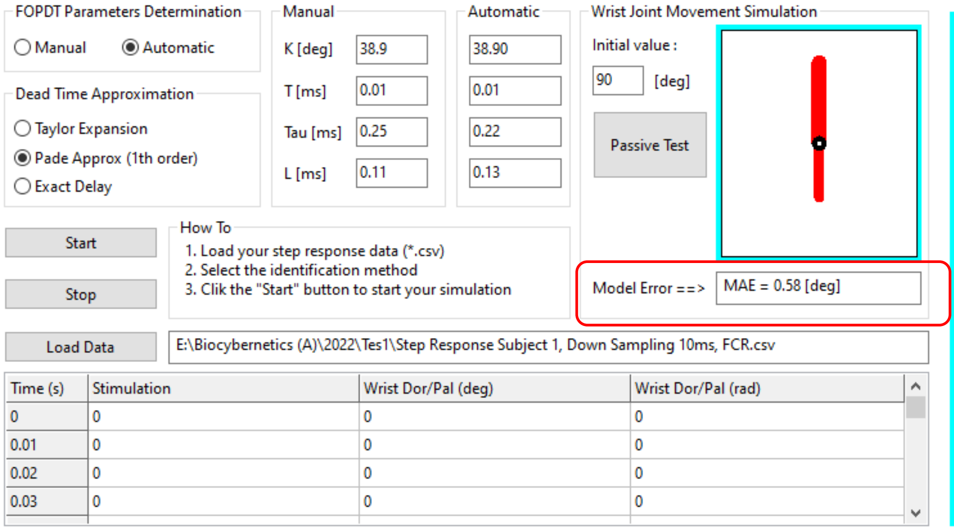
$$G(s) = \frac{Ke^{-Ls}}{\tau s + 1} = \left(\frac{K}{\tau s + 1}\right)\frac{2 - Ls}{2 + Ls}$$

$$G(s) = \left(\frac{K}{\tau s + 1}\right)\frac{2 - Ls}{2 + Ls} \rightarrow G(z) = G(s)\Big|_{s = \frac{2}{T}\frac{z-1}{z+1}}$$

$$G(z) = \frac{a + (a+b)z^{-1} + bz^{-2}}{c + (e-d)z^{-1} - fz^{-2}}$$

$$a = KT(T-L),\ b = KT(T+L)$$
$$c = (T+L)(2\tau + T), d = (T+L)(2\tau - T)$$
$$e = (T-L)(2\tau + T),\ f = (T-L)(2\tau - T)$$

$$y(n) = \frac{1}{c}[(d-e)y(n-1) + fy(n-2) + au(n) + (a+b)u(n-1) + bu(n-2)]$$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K [deg] 38.9
T [ms] 0.01
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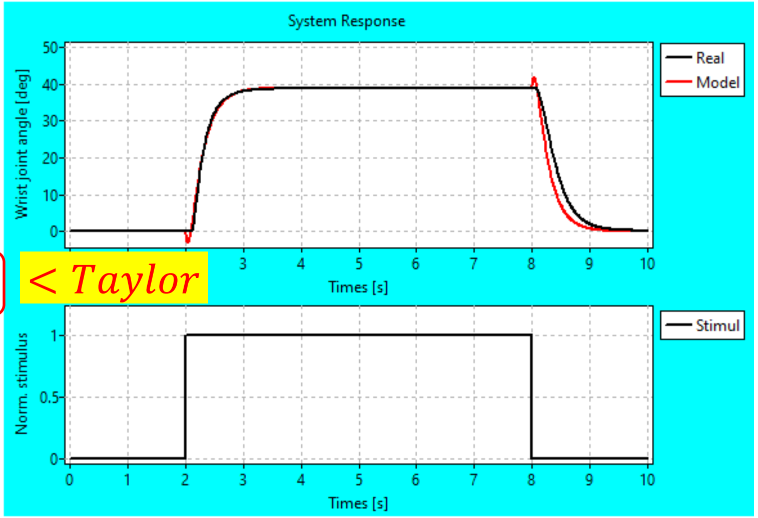
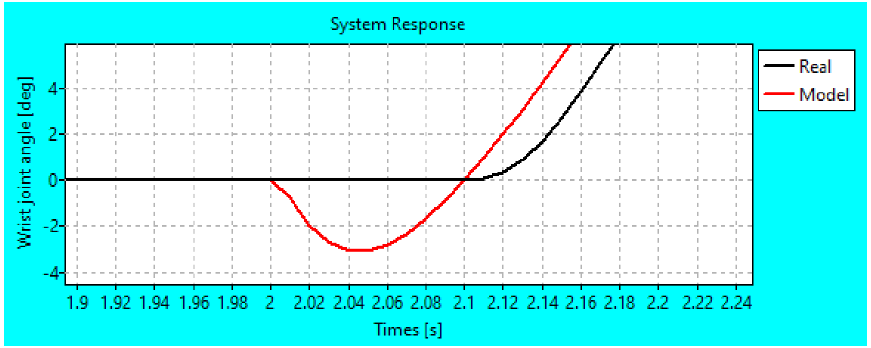
L [ms] 0.11
Automatic
38.90
0.01
0.22
0.13
Wrist Joint Movement Simulation
Initial value :
90 [deg]
Passive Test
Start
Stop
How To
1. Load your step response data (*.csv)
2. Select the identification method
3. Clik the "Start" button to start your simulation
Model Error ==> MAE = 0.58 [deg]
Load Data
E:\Biocybernetics (A)\2022\Tes1\Step Response Subject 1, Down Sampling 10ms, FCR.csv
Time (s) | Stimulation | Wrist Dor/Pal (deg) | Wrist Dor/Pal (rad)
0 | 0 | 0 | 0
0.01 | 0 | 0 | 0
0.02 | 0 | 0 | 0
0.03 | 0 | 0 | 0


System Response
Wrist joint angle [deg]
Real
Model
< Taylor
Times [s]
Norm. stimulus
Stimul
Times [s]


System Response
Wrist joint angle [deg]
Real
Model
Times [s]

3. Direct transformation (Exact delay): $z = e^{sT}$

$$e^{-sL} = z^{-L/T}$$

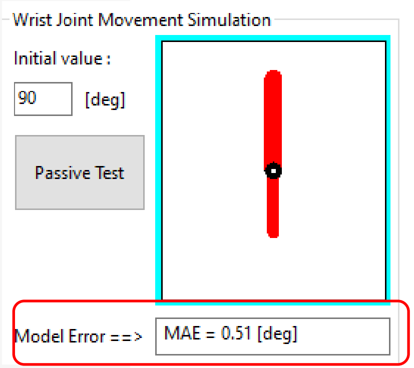
Bilinear transformation:

$$G(s) = \frac{Ke^{-Ls}}{\tau s + 1} \rightarrow G(z) = G(s)\Big|_{s = \frac{2}{T}\frac{z-1}{z+1}}$$

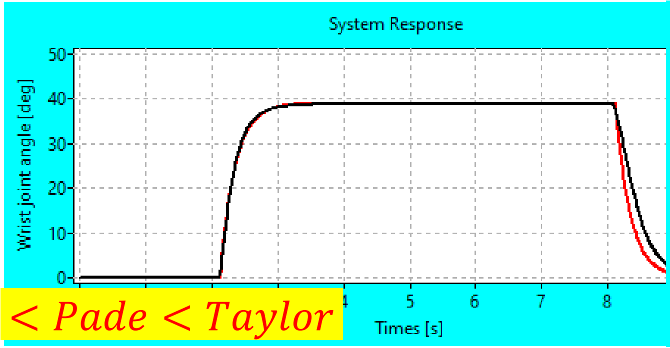
$$G(z) = \frac{Kz^{-L/T}}{\left(\frac{2\tau}{T}\frac{1 - z^{-1}}{1 + z^{-1}}\right) + 1}$$

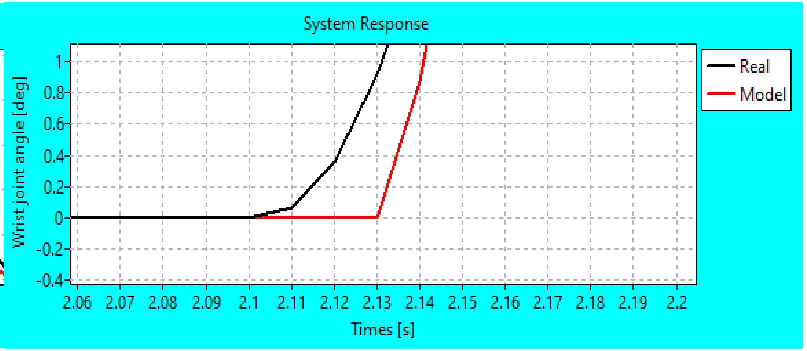
$$y(n) = -ay(n-1) + b\left[u\left(n - \frac{L}{T}\right) + u\left(n - \frac{L}{T} - 1\right)\right]$$

$$a = \frac{(T - 2\tau)}{(T + 2\tau)} \qquad b = \frac{KT}{(T + 2\tau)}$$

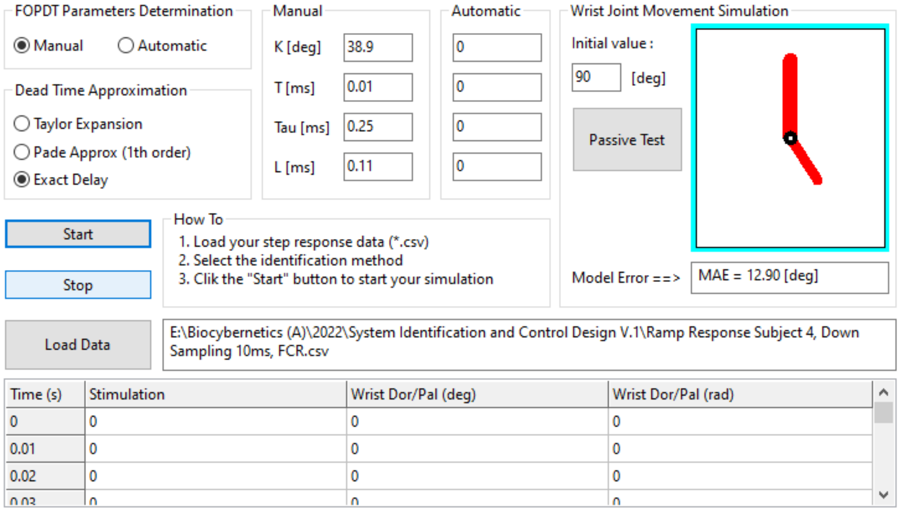


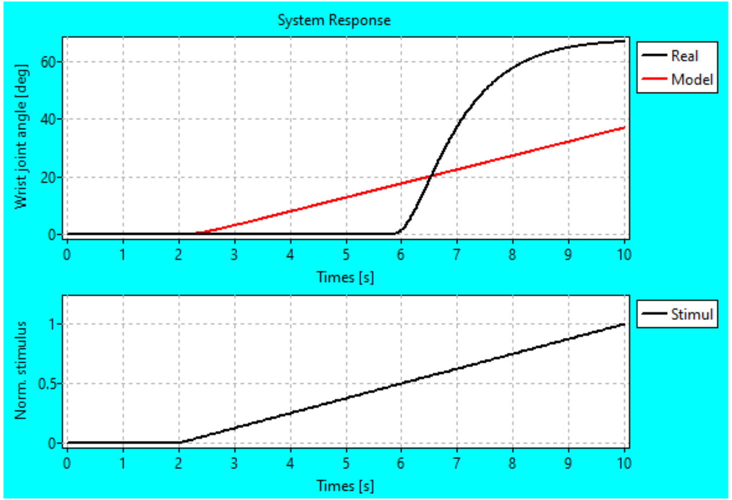
$< Pade < Taylor$





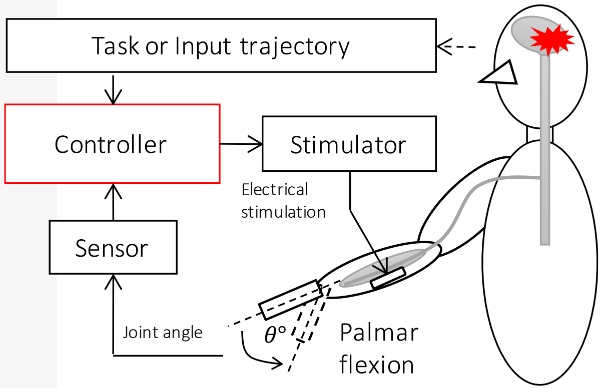
Approximation Effect:



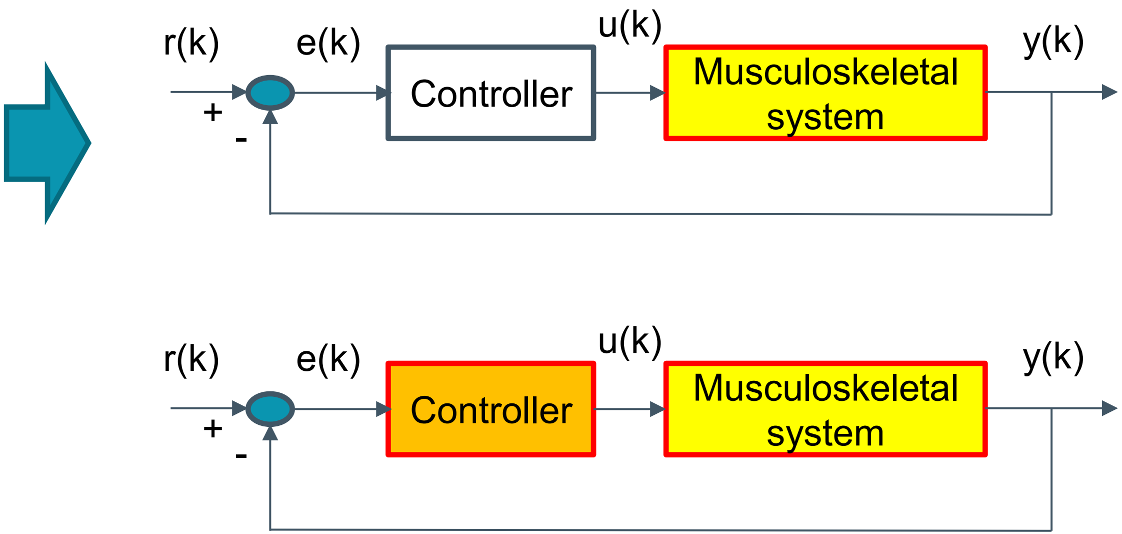


FES Controller for Wrist Joint Control (Computer Simulation)

Closed-Loop FES Control System



Computer Simulation



Control System Design

1. Conventional Feedback (Error-Based) Controller
   - PID Controller
   - Fuzzy Controller
2. Model-Based Controller
   - Hybrid Controller → Feedforward + Feedback Controller
   - Model Predictive Control (MPC) → optimal control technique